

Optimizing Profitability & Growth

Jaffle Shop Analytics Engineering Hackathon

July 10th

The Dual Strategy

Moving beyond raw order volume to uncover true store efficiency and untapped revenue synergies.

Analytics **Architecture**

- **Staging Layer:** Cleaned and mapped raw CSVs for Orders, Items, Products, Supplies, and Stores.
- **Marts Layer:** Developed two highly tested fact tables: `fct_store_margins` and `fct_product_affinity`.
- **Testing rigor:** Enforced unique primary keys, not-null constraints, and strict business logic guardrails.

600 x

MarginMaker: **Store Efficiency**

- > **The Question:** Which stores are losing money on high supply costs?
- > **The Engineering:** We joined item-level revenue against localized ingredient/supply costs to find net profit.
- > **Business Logic Test:** We built a custom dbt test to ensure profit margins never register as mathematically impossible negative values.

600 × 400

The Insight & Action

+18%

PROFIT VARIANCE

Efficiency over Volume

Insight: Store A processes fewer total orders than Store B, yet maintains an 18% higher profit margin due to superior supply-cost controls.

Actionable Next Step: Execute an immediate audit of Store B's local inventory waste and mandate a transition to Store A's supplier network.

BasketWeaver: Cross-Selling

- > **The Question:** Which products organically drive the sales of others?
- > **The Engineering:** We utilized a complex self-joining CTE to execute a market basket analysis across all historical orders.
- > **Business Logic Test:** We tested that ``product_a` != `product_b`` to guarantee our pipeline never artificially pairs a product with itself.

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The Insight & Action

35%

CO-OCCURRENCE RATE

Unlocking Free Revenue

Insight: Product X and Product Y are naturally purchased together in 35% of all orders where either item is present.

Actionable Next Step: Redesign the digital menu and POS system to offer a '1-Click Combo' upsell, instantly increasing Average Order Value (AOV).

Phase 2: **Production & Beyond**



Predictive ML Store

Leveraging our fact tables as a clean feature store for BigQuery ML clustering, predictive margin forecasting, and recommendation engines.



Incremental Builds

Upgrading model materializations to incremental to process only new data, vastly optimizing cloud compute costs at scale.



CI/CD Orchestration

Implementing GitHub Actions for automated testing on PRs and scheduling DAG runs via Apache Airflow to guarantee data freshness.

Questions?

Ready to deploy these models to production.

 `github.com/your-username/jaffle-shop-dbt`

Image Sources

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Source: www.vecteezy.com

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<https://www.slidekit.com/wp-content/uploads/2026/06/Profit-Loss-Dashboard-Template.jpg>

Source: www.slidekit.com

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